Miscellaneous Theorems and Ideas

Jonathan Kasongo

January 8, 2026

This is just a collection of some cool facts and theorems I have found, along with their proofs.

Theorem 1.

For an odd positive integer n, and positive integers a, b with $a^2 \leq b < (a+1)^2$ we have the following,

$$\left\lfloor \left(a+\sqrt{b}\right)^n \right\rfloor = \left(a+\sqrt{b}\right)^n + \left(a-\sqrt{b}\right)^n.$$

Proof of Theorem 1.

Firstly, due to Binomial expansion,

$$\begin{aligned}
\left(a+\sqrt{b}\right)^n + \left(a-\sqrt{b}\right)^n &= \sum_{r=0}^{n} \binom{n}{r} \left(\sqrt{b}\right)^r a^{n-r} + \sum_{r=0}^{n} \binom{n}{r} \left(-\sqrt{b}\right)^r a^{n-r} \\
&= \sum_{r=0}^{n} \binom{n}{r} \left(\sqrt{b}\right)^r [1+(-1)^r]\, a^{n-r} \\
&= 2 \sum_{\substack{r=0 \\ r \text{ even}}}^{n} \binom{n}{r} \left(\sqrt{b}\right)^r a^{n-r}
\end{aligned}$$

which is clearly an integer. Now it suffices to show that

$$\left(a+\sqrt{b}\right)^n - 1 < \left(a+\sqrt{b}\right)^n + \left(a-\sqrt{b}\right)^n \leq \left(a+\sqrt{b}\right)^n$$

That is

$$-1 < \left(a-\sqrt{b}\right)^n \leq 0$$

but we know that $a \leq \sqrt{b} < a+1$ and so $a-\sqrt{b} \leq 0$ and $-1 < a-\sqrt{b}$ and raising both inequalites to the power of n gives the desired result. ■

Theorem 2.

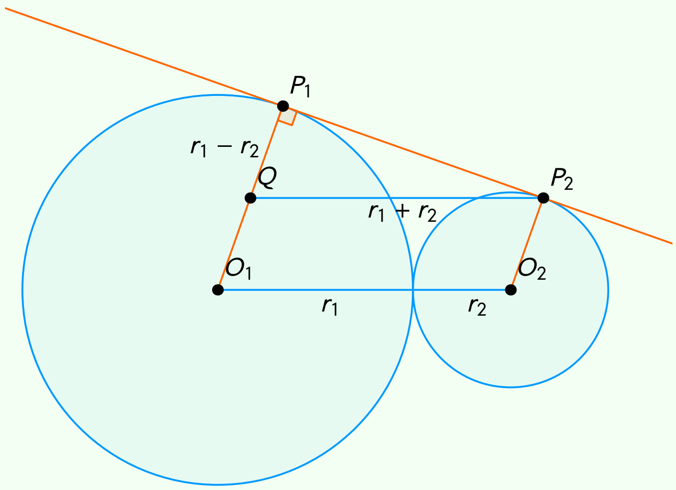
Let C_1 and C_2 be externally tangent circles with centres O_1, O_2 and radii $r_1, r_2, (r_1 \geq r_2)$ respectively. Let P_1, P_2 be points on C_1, C_2 respectively such that P_1P_2 is a common tangent to both circles.

(a) Show that $P_1P_2 = 2\sqrt{r_1r_2}$.

(b) Find the area of the region bounded by the two circles and their common tangent.

Solution

(a) The idea is to draw in the line l through P_2 that is parallel to O_1O_2. Let $Q = l \cap O_1P_1$.



We notice that since O_1P_1 and O_2P_2 are radii they meet the common tangent at right angles, and in particular $O_1P_1 \parallel O_2P_2$. Now it is clear to see that $QP_2 = r_1 + r_2$ and $QP_1 = r_1 - r_2$. Now owing to Pythagoras,

$$P_1P_2 = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1r_2}. \square$$

(b) Notice that due to the parallel conditions $O_1P_1P_2O_2$ is a trapezium, with area

$$\frac{r_1 + r_2}{2}\left(2\sqrt{r_1r_2}\right).$$

Now we just need to subtract off the two sectors. Let $\angle P_1O_1O_2 = \theta$. Notice that $\angle O_1O_2P_2 = 180 - \theta$ due to corresponding angles. Taking θ to be measured in radians gives the area of the two sectors to be

$$\frac{1}{2}r_1^2\theta + \frac{1}{2}r_2^2(\pi - \theta) = \frac{1}{2}\theta\left(r_1^2 - r_2^2\right) + \frac{1}{2}\pi r_2^2.$$

So the final area is

$$(r_1 + r_2)\sqrt{r_1r_2} - \frac{1}{2}\theta\left(r_1^2 - r_2^2\right) + \frac{1}{2}\pi r_2^2. \blacksquare$$

Comments: I first came across the fact (a) from a BPRP (Black pen red pen) video, and then I actually used this fact to solve a problem in a Year 12/13 geometry topic test. Part (b) actually came up as a question with special case $r_1 = 3, r_2 = 1$ when I was sitting the 2026 Jan TMUA, I was unable to solve it during contest.